Gamma-Ray Spectrometer

ANALOGICS
Redefining Technologies...

- Indigenously Developd product
- Display of K, U, Th concentrations
- Data acquisition
- Rugged design for field use
- Low cost and versatile
- Software for data acquisition, interface, storage, transfer of data

Features:

- Display of K, U, Th concentrations, dose rate with a user selectable sample time (Acquisition time)
- Data Logging
- Rugged Design to withstand field usage
- Dynamic range: 10 to 3000 KeV
- Resolution: < 7%, FWHM at 662 KeV, <5% at 1.33 MeV
- Count rate: 50,000 cps
- Supports data acquisition, transfer of data, interface and storage
- Offline software supports Spectrum processing
- Digital potentiometer for calibration

Multi Channel Analyzer:

256/512/1024 Channel MCA
Integral non linearity: <0.07%
Differential non linearity: <+/-1%
Dead time for event: 2usec including memory transfer
Gain stability: <+50ppm/degC

Data Acquisition System:

ARM controller based DAS with Linux OS
Data acquisition from the MCA and GPS
Mode selection and Display
High voltage and amplifier gain
Built in GPS

Application Areas

Monitoring Airports, Shipping Yards & Nuclear Facilities
Radiation Detection
Toxic dump site monitoring
Atomic Research
Awareness in Educational Institutions

Embedded Software:

-> Acquisition and display of the live radiometric data at an interval of 5 seconds
Control of amplifier gain with digital potentiometer
Computation and display of dose rate, radio elemental concentration after the user defined acquisition time
Storage of more than 500 sample spectrum along with GPS coordinates Transfer of the stored data to PC/ Server

Specifications:

Processor and OS	:	High speed ARM Cortex A8 processor with 1GHz Linux OS Kernel 3.2
Memory	:	512 MB RAM 4 GB Flash
Interface	:	USB 2.0 Host, USB 2.0 Client, GPS Module (Internal)
Detector	:	Standard detector 3” x 3” NaI(Tl) with PMT
Power	:	Rechargeable Li-Ion Battery, 2600mAh capacity, for >8 hours operation
Display and Keypad	:	3.5” QVGA high bright TFT with Touch Panel Panel mounted keypad with 5 keys (Up, Down, Calibration, Menu, Ok), 54 pin connector for Display
Indications	:	Charge LED and Power ON LED
Operating Temperature	:	0 – 50 °C
Storage Temperature	:	Auto calibration with Cesium (Cs) & Potassium(K) Built in GPS RTC (Real Time Clock) Wifi Audio
Dimensions	:	260mm (L) x 100mm (W) x 110mm (H)
Weight	:	Less than 3 kg

Note:

Option 1 : Single unit integrated of Electronic Device & Sensor with PMT.

Option 2 :Separate Electronic Device & Sensor with PMT will be interconnected through cable.

Product Description

Indigenously developed product to display K, U and Th concentrations. It is a low cost product and compatible for field use due to rugged design. The Software supports Data Acquisition, Transfer of Data, Interface and Storage.

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Product features and specifications are subject to change without notice to improve reliability, performance, design or otherwise. Please refer to the website for latest updates